CSE225 Data Structures **Fall 2011 21/10/2011** **Student Name:_____**
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Q1. (10+15 = 25 pts)
Consider the C function below!

```
func (int n, long count){
        int i;
        if (n<2) {count=n;   return;}
        for (i=1; i<=n; i++)  count ++;
        func(n-2, count);
        return;
}
```

a. Specify the recurrence and initial values that represents the execution time *t(n)* of func!
b. Find the particular solution for the above recurrence!

a. $t(n) = t(n-2) + n$ $t(0)=0,\ t(1)=0$

b. CE: $(x^2-1)(x-1)^2 = 0$
The general solution:
$t(n) = c_1 + c_2 n + c_3 n^2 + c_4(-1)^n$
$c_1 = -9/8$ $c_2 = 5/2$ $c_3 = -1/4$ $c_4 = 9/8$

Q2. (5+5+5= 15 pts)
Solve the following recurrences!

a. $T(n) - 2T(n/2) = n + \log_2 n;$ $T(1) = 0$
b. $T(n) - 2T(n-1) = (n+5)3^n;$ $T(0) = 3$

a. Assume $n=2^k$, $T(k) - 2T(k-1) = 2^k+k$
CE: $(x-2)(x-2)(x-1)^2$
The general solution:
$T(k) = c_1 2^k + c_2 k 2^k + c_3 + c_4 k$ SO;
$T(n) = c_1 n + c_2 \log_2 n * n + c_3 + c_4 \log_2 n$

$T(1) = 0 = c_1 + c_3$
$T(2) = 2T(1)+2+1= 3 = 2c_1 + 2c_2 + c_3 + c_4$
$T(4) = 2T(2)+4+2=12 = 4c_1 + 8c_2 + c_3 + 2c_4$
$T(8) = 2T(4)+8+3=35 = 8c_1 + 24c_2 + c_3 + 3c_4$
$c_1 = 2$ $c_2 = 1$ $c_3 = -2$ $c_4 = -1$ => $T(n) = 2n + \log_2 n*(n-1) - 2$

b. CE: $(x-2)(x-3)^2 = 0$
The general solution:
$t(n) = c_1 2^n + c_2 3^n + c_3 n 3^n$
$T(0) = 3 = c_1 + c_2$
$T(1) = 2T(0)+18= 24 = 2c_1 + 3c_2 + 3c_3$
$T(2) = 2T(1)+63=111 = 4c_1 + 9c_2 + 18c_3$

$c_1 = -6$ $c_2 = 9$ $c_3 = 3$ => $T(n) = -6*2^n + 9*3^n + 9*n3^n$